Causal Regime-Conditioned Trading Agent: Verifiable Regime-Aware Reinforcement Learning

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***Abstract*—Regime-aware reinforcement-learning (RL) trading agents are usually judged by back-tested returns, which cannot show whether the regime signal drives their decisions. We pair a Wasserstein regime-identity tracker for walk-forward hidden-Markov-model refits with three interventions on the regime input, and compare a proximal policy optimisation (PPO) agent with a twin that lacks that input. On BTC/USDT four-hour bars the tracker keeps a stable volatility ordering across 292 causal refits and lowers the mean template-matching cost by a factor of 4.9, but it starves the bull template and issues no bull label after burn-in. Ablation moves the conditioned position on 60.7% of 11,430 out-of-fold bars, yet only 0.1% of bars pass all three checks (0.25% of the 4,390 bars on which the position limit allows the counterfactual move), against 33.5% for moment-sorted labels. Repeating that study on five seeds, the conditioned mean return of $+57.8\%$ is one seed at $+515\%$; the other four lose and the mean Sharpe ratio is -0.27. Five seeds of timeframe-native twins on BTC/USDT and ETH/USDT at 15 min and 1 h, with the recipe frozen on validation, give a negative mean holdout Sharpe ratio for both twins, and the conditioned twin beats the blind twin on a majority of seeds in two of eight windows only. Those twins hold $|a| > 0.95$ on 97–100% of bars. Validation screens of a tanh head, a five-level discrete head and a mean-variance reward, alone and combined, found no setting that moves the policy off the rails while keeping it responsive.**

***Index Terms*—regime detection, hidden Markov model, Wasserstein distance, reinforcement learning, algorithmic trading, explainability, faithfulness, counterfactual testing**

I. Introduction

Cryptocurrency markets move through very different conditions within a few years. Bitcoin rose sharply in 2021, lost roughly three quarters of its value between late 2021 and late 2022, and then spent 2023–2026 alternating between recoveries and drawdowns. A natural remedy for such non-stationarity is to detect the prevailing regime and hand it to the policy as an extra input. Hidden-Markov-model (HMM) regimes have been fed to RL portfolio agents with stronger reported risk-adjusted returns [1], and regimes have been used for sector portfolios [2] and asset allocation [3].

What this work rarely checks is whether the regime input matters. An agent can report that it "cut exposure because a crisis was detected" while its computation ignores the crisis probability or responds to it in the wrong direction. An explanation that is easy to understand is not necessarily faithful to the computation behind it [4]. A second problem sits inside the detector. An HMM refitted on new data returns its states in arbitrary order. Sorting states by their moments after every fit forces four names onto whatever four components the latest fit produces. We instead keep persistent bull, bear, calm and crisis templates and match each new component to the nearest template by the closed-form 2-Wasserstein distance, so a name survives a refit only when the component still resembles it.

We ask whether Wasserstein tracking gives labels that keep their meaning across walk-forward refits, and whether an RL agent conditioned on them depends on them in the way their names imply. Our contributions are:

- A walk-forward Wasserstein template tracker for a four-state Gaussian HMM, with matching, aggregation and template smoothing written as equations that match the released code.
- A per-decision verification suite of three interventions on the regime input, validated on reference policies that are faithful, regime-ignoring and inverted by construction, and a twin experiment in which conditioned and blind PPO agents differ only by that input.
- A four-hour BTC/USDT study in which the regime input is load-bearing but the faithfulness rate collapses from 33.5% to 0.1%, traced to template starvation and out-of-distribution counterfactuals, with a five-seed repetition showing that its mean return is one seed.
- A five-seed study of timeframe-native twins on BTC/USDT and ETH/USDT at 15 min and 1 h with a recipe frozen on validation, and a negative result on the position rails: neither other action heads nor a mean-variance reward produced a policy that uses the interior of $[-1, 1]$ while staying responsive.

II. Related Work

Regimes and RL for portfolios. Ndoutoumou *et al.* pass a Gaussian-HMM regime into RL agents on a Dow Jones portfolio and report better Sharpe and Sortino ratios than the same agents without it [1]; the comparison is on back-tested returns and does not test whether the policy uses the regime. Regimes have also been detected with directional-change indicators in foreign exchange [5], classified with explainable models for asset allocation [3], and used for sectoral portfolios in non-stationary markets [2]. Deep RL for

portfolio choice is surveyed in [6], [7] and applied to asset-weight management [8], dynamic capital allocation [9], real-time rebalancing with PPO and DDPG [10], single-stock portfolios [11], A-share portfolios under trading constraints [12] and adversarial market conditions [13]. Across this work a higher Sharpe ratio is treated as evidence that the agent has adapted. None of it checks whether a named regime input is what the policy uses.

Explainability and causality. Mamalakis *et al.* separate a faithful explanation from a merely comprehensible one [4]. Wiratsin and Ragkhitwetsagul note the lack of standard evaluation metrics for explainable AI [14], Yeo *et al.* survey post-hoc explanations in finance [15], and Reinhardt and Buchholz argue that explainability in high-stakes domains is an obligation [16]. Causal models of the market have been built for stock-trend forecasting [17] and for decisions in changing environments [18]. Our tests differ in kind: the policy is a known function, so setting one named input measures that input's effect on the action, not a causal effect in the market.

III. Preliminaries

Let $x_t \in \mathbb{R}^3$ be the regime features at bar t. A K-state Gaussian HMM has latent states z_t, a transition matrix A and emissions $x_t \mid z_t = k \sim \mathcal{N}(m_k, \mathrm{diag}(v_k))$, fitted by expectation–maximisation (EM). The posterior $\gamma_t(k) = P(z_t = k \mid x_{1:n})$ comes from the forward–backward recursion and the most likely path from Viterbi decoding. The likelihood is invariant to permuting state indices, so two fits on overlapping data can return the same regimes under different indices.

For Gaussians the 2-Wasserstein distance has a closed form. With diagonal covariances $\Sigma_i = \mathrm{diag}(s_i^2)$ the covariance term reduces to $\|s_1 - s_2\|^2$, so

$$W_2^2\big(\mathcal{N}(m_1,\Sigma_1),\mathcal{N}(m_2,\Sigma_2)\big) = \|m_1 - m_2\|^2 + \|s_1 - s_2\|^2. \tag{1}$$

It compares location and spread on the feature scale at $O(d)$ cost.

A trained deterministic policy $a = \pi_\theta(o)$ is a known function of its observation $o = (u, p)$, where p is the regime-probability block. We write $\pi_\theta(u, \mathrm{do}(p{=}p'))$ for the action when p is set to p' and everything else is held fixed. Because π_θ is available in closed form, this needs no causal identification assumptions. A regime signal is *load-bearing* if such interventions change the action, and *faithful to its names* if the change has the sign the names imply.

IV. Proposed Methodology

Fig. 1 shows the pipeline. All code is in the `crta` package of the project repository.

A. Causal features and walk-forward refits

Let C_t be the close of bar t. The features are the log return $r_t = \log(C_t/C_{t-1})$, the 20-bar rolling standard deviation σ_t of r, the trend $\tau_t = C_t/\mathrm{SMA}_{20}(C)_t - 1$, the 14-bar Wilder RSI scaled to $[0,1]$, and the MACD histogram divided by C_t. Each feature at t uses rows $\le t$ only, and a unit test checks that features on a truncated history equal the prefix of the full computation. The regime model uses $x_t = (r_t, \sigma_t, \tau_t)$. The agent sees all five as f_t, rescaled to unit order by a fixed map ϕ.

The tracker makes one forward pass over the series (Fig. 2). The first $B = 252$ bars (six weeks) are a burn-in. Refit j starts at $t_j = B + jS$ with stride $S = 42$ bars (one week). A four-state diagonal HMM is fitted by EM (200 iterations, seed 42) to the rows strictly before t_j, standardised with that window's mean $\bar{x}$ and standard deviation s, and decodes the window $W_j = \{t_j, \ldots, t_j + S - 1\}$. Component parameters are mapped back to feature units, $\mu_k = \hat{m}_k \odot s + \bar{x}$ and $\sigma_k = \sqrt{\hat{v}_k} \odot s$. The burn-in fit is the only place where names come from moments: of the two components with the lowest mean return, the less volatile is *bear* and the other *crisis*; of the two highest, the less volatile is *calm* and the other *bull*. This yields templates $(\bar{\mu}_g, \bar{\sigma}_g)$ for $g \in \mathcal{G} = \{\text{bull}, \text{bear}, \text{calm}, \text{crisis}\}$.

B. Wasserstein identity matching and template update

At refit j the cost of naming component k as regime g is the diagonal distance (1), $D_{kg} = \|\mu_k - \bar{\mu}_g\|^2 + \|\sigma_k - \bar{\sigma}_g\|^2$, and every component takes its nearest template,

$$\pi_j(k) = \arg\min_{g \in \mathcal{G}} D_{kg}. \tag{2}$$

The matching need not be one-to-one. The regime probabilities for $t \in W_j$ are the summed posteriors

$$p_t(g) = \sum_{k:\, \pi_j(k) = g} \gamma_t(k), \tag{3}$$

and an unmatched template has $p_t(g) = 0$. After W_j is labelled, every matched template moves a step $\eta = 0.2$ toward its members. With $w_k = \sum_{t \in W_j} \gamma_t(k)$, member set $M_g = \{k : \pi_j(k) = g\}$ and $\omega_k = w_k / \sum_{k' \in M_g} w_{k'}$,

$$\bar{\mu}_g \leftarrow (1 - \eta)\, \bar{\mu}_g + \eta \sum_{k \in M_g} \omega_k \mu_k, \tag{4}$$

and likewise for $\bar{\sigma}_g$. A template with $M_g = \emptyset$ is left unchanged. The hard label $\hat{h}_t = \pi_j(\hat{z}_t)$ names the Viterbi state; runs shorter than $L = 30$ bars (five days) are merged into the preceding label. Hard labels are used only for reporting. The agent sees p_t.

C. Twin agents

At bar t the agent chooses a target position $a_t \in [-1, 1]$ and receives

$$R_t = a_t\, r_{t+1} - c\,|a_t - a_{t-1}|, \qquad c = 10^{-3}, \tag{5}$$

the next log return less a 10-basis-point turnover cost. The conditioned agent observes $o_t = [\phi(f_t), a_{t-1}, p_t] \in \mathbb{R}^{10}$ and the blind twin $[\phi(f_t), a_{t-1}] \in \mathbb{R}^6$. Both are Stable-Baselines3 PPO agents with identical settings; the only architectural difference is four extra input weights.

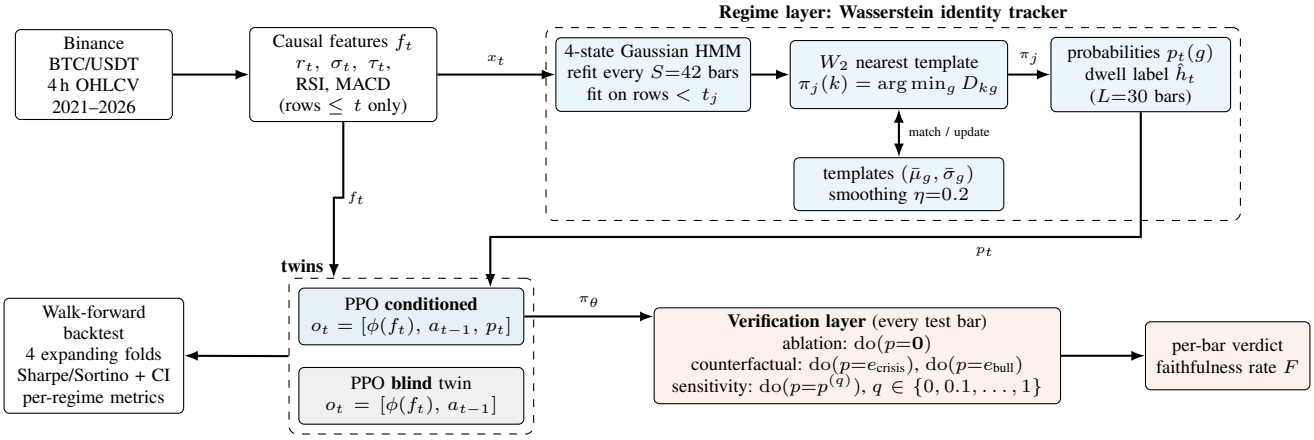


Fig. 1. System architecture. Features are causal. The regime layer refits the HMM on past rows only, names its components by Wasserstein distance to persistent templates, and passes the aggregated probabilities p_t to the conditioned agent. The blind twin differs only in lacking p_t. Every out-of-fold decision of the conditioned agent is audited by three interventions on p_t.

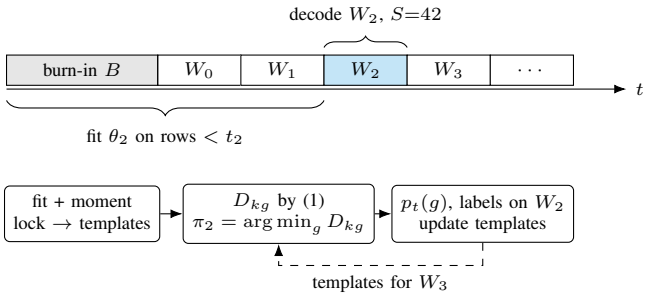


Fig. 2. Causal refit schedule. The burn-in fit names the templates once by moment sorting. Every later window is decoded by a model fitted only on earlier rows, and its components inherit names by Wasserstein matching. Labels already issued are never revised.

D. Verification suite

For each out-of-fold bar we freeze the policy, the features and the previous position, and intervene on p alone. Let $a_t = \pi_\theta(u_t, p_t)$, e_g the one-hot vector of regime g, and ρ a defensive rank with $\rho(\text{bull}) = 0 < \rho(\text{calm}) = 1 < \rho(\text{bear}) = 2 < \rho(\text{crisis}) = 3$.

Ablation. The regime input is load-bearing at t if

$$\Delta_t^{\text{ab}} = \left| a_t - \pi_\theta(u_t, \text{do}(p{=}\mathbf{0})) \right| \ \geq\ \delta_{\text{ab}} = 0.15. \tag{6}$$

Counterfactual. With $g_t = \arg\max_g p_t(g)$, forcing crisis must cut the position and forcing bull must raise it, each by at least $\delta_{\text{cf}} = 0.10$:

$$\rho(g_t) < 3 \Rightarrow a_t - \pi_\theta(u_t, \text{do}(p{=}e_{\text{crisis}})) \geq \delta_{\text{cf}}, \tag{7}$$

$$\rho(g_t) > 0 \Rightarrow \pi_\theta(u_t, \text{do}(p{=}e_{\text{bull}})) - a_t \geq \delta_{\text{cf}}. \tag{8}$$

Sensitivity. For $q \in \{0, 0.1, \ldots, 1\}$ we inject a crisis probability q and rescale the rest,

$$p^{(q)} = (1 - q)\, \frac{p_t^{\setminus c}}{\|p_t^{\setminus c}\|_1} + q\, e_{\text{crisis}}, \tag{9}$$

where $p_t^{\setminus c}$ is p_t with its crisis entry set to zero. The curve $y_i = \pi_\theta(u_t, \text{do}(p{=}p^{(q_i)}))$ passes if it rises at most once.

A bar is *faithful* when all three checks pass, and the faithfulness rate F is the share of faithful bars. *Verifier v2* replaces the one-hot vectors by the nearest training-set probability vectors whose argmax is crisis or bull, rejects a flat sensitivity curve, and marks a bar untestable rather than failed when the training set has no such vector.

V. Experimental Setup

A. Four-hour study

We use Binance BTC/USDT four-hour candles from 1 Jan 2021 to 22 Sep 2026 (12,544 bars; 12,510 after feature warm-up). Four expanding walk-forward folds start with 1,080 training bars, and each test block of about 2,857 bars is scored by agents trained only on the bars before it (Table I). The test blocks give 11,430 out-of-fold bars from 5 Jul 2021 to 22 Sep 2026. Labels come from the single causal tracker pass.

Settings. HMMs use `hmmlearn` with minimum covariance 10^{-3}. The policy is an MLP $64{-}64$ with tanh units and a Gaussian action head clipped to $[-1, 1]$. PPO uses learning rate $3{\times}10^{-4}$, γ=0.99, λ_{GAE}=0.95, entropy coefficient 0, n_{steps}=256, batch 64, clip 0.2, 10 epochs, seed 42, and $\max(8192, 10\, N_{\text{train}})$ steps per agent (ten passes over the training window). The full pipeline (292 refits, eight PPO trainings and the audit) ran in 6.6 minutes on a laptop CPU, and all 32 unit tests pass, including feature leakage, the closed-form distance and label stability under a component swap.

Baselines. The blind twin (Sec. IV-C) and buy-and-hold ($a_t = 1$). The *archived moment-sort run* is the pipeline before the tracker was introduced: one HMM per fold named by moments alone, everything else identical. Its blind-twin and buy-and-hold results match ours exactly, so the labels are the only change. For label statistics we also re-name all 292 refits by moment sorting instead of (2). The verifier is validated on three reference policies over (bull, bear, calm, crisis): $a = \text{clip}(p^\top w)$ with $w = (1, -0.4, 0.5, -1)$ is

TABLE I
FOUR-HOUR WALK-FORWARD FOLDS AND PER-FOLD OUT-OF-FOLD SHARPE RATIOS (ANNUALISED)

Fold	Test period	Train	Cond.	Blind	B&H	Cond.†
0	07/2021–10/2022	1,080	−1.95	−1.84	−0.65	−2.32
1	10/2022–02/2024	3,937	0.24	−3.29	1.69	−0.53
2	02/2024–06/2025	6,794	0.53	0.66	1.10	2.24
3	06/2025–09/2026	9,651	0.50	−0.79	−0.39	1.93

†Archived run with per-fold moment-sorted labels (Sec. V-A).

faithful by construction, $a = 0.5$ ignores the regime, and $a = \mathrm{clip}(-p^\top w)$ inverts it.

Metrics. We report the annualised Sharpe ratio $\sqrt{2190}\,\bar{R}/\hat{s}_R$ and the Sortino ratio with 95% percentile intervals from 1,000 i.i.d. bootstrap resamples, maximum drawdown and total return, each also within each dwell-smoothed regime. For the tracker we report label shares, switches per year, run lengths and the mean matched W_2^2 cost.

B. Timeframe-native twins

The 15-min and 1-h tapes of BTC/USDT and ETH/USDT get twins trained on their own candles: 202,596 bars on 15 min and 50,627 on 1 h per asset after warm-up. The tracker keeps the calendar spans of the four-hour study, $(B, S, L) = (1{,}008, 168, 120)$ on 1 h and $(4{,}032, 672, 480)$ on 15 min, in one pass per series. A single episode over the full training window sent both BTC 1-h twins towards a constant long position, so each episode is a randomly placed stretch of 2,000 training bars started flat, the reward (5) is multiplied by 100 during training only, and each twin trains for 500,000 steps.

Splits. For validation, twins trained on bars before 22 Sep 2024 16:00 UTC are scored on the following year. For the holdout, twins retrained on bars before 22 Sep 2025 16:00 UTC are scored on the year up to the live window. The live window scores every candle from 22 Sep 2026 16:00 UTC to a snapshot on 13 Oct 2026 at 10:47 UTC (1,994 decisions on 15 min and 497 on 1 h, 21 days) with the same frozen weights. Native runs use seeds 42–46. We report the mean and the standard deviation across seeds; Sharpe ratios use 35,040 and 8,760 bars per year. "Beats" is the share of seeds whose conditioned log-return sum exceeds the blind twin's.

Live console. A Streamlit console runs both twins on paper money as Binance candles close and never sends an order. Only closed candles reach an agent, a failed request leaves the stored candles untouched, and sliders apply $\mathrm{do}(p{=}p')$ to the conditioned agent so the interventions can be explored by hand. Its start-up labelling is faster than the tracker; every live number reported here comes from the frozen study and the tracker.

VI. RESULTS AND DISCUSSION

Tables I–IV and Figs. 3–7 are the published single four-hour run. They were not recomputed; Section VI-D repeats that recipe on five seeds beside them.

TABLE II
FOUR-HOUR REGIME LABELS AFTER BURN-IN (12,258 BARS): SHARE OF BARS, MEAN 4 H LOG RETURN AND MEAN σ_{20} PER LABEL

	W_2 tracker (ours)			Moment sort per refit		
Label	share	ret (bp)	σ (%)	share	ret (bp)	σ (%)
bull	0.0%	–	–	7.2%	−0.8	1.54
bear	7.7%	−7.7	1.42	34.2%	−3.1	0.93
calm	88.5%	+1.5	0.97	44.2%	+3.7	1.04
crisis	3.8%	−8.1	2.18	14.4%	−0.6	1.14
switches / yr		5.5			7.5	
median run (bars)		110			210	
mean matched W_2^2 (10^{-4})		1.88			9.13	

A. Regime identity under refits

Fig. 3 shows the price with the tracker's labels and both labellings, and Table II compares the two labellings on the same refits. Moment sorting assigns a different name than the W_2 rule to at least one component in 291 of the 292 refits, and to 49.6% of all components. Its labels are semantically unstable: its *bull* bars have a negative mean return (−0.8 bp), and its *crisis* bars are only slightly more volatile than its *calm* bars (1.14% against 1.04%). The W_2 tracker keeps the ordering the names promise: crisis bars average 2.18% volatility and −8.1 bp per bar, against 0.97% and +1.5 bp for calm. It switches less often (5.5 against 7.5 times per year), its matched cost is 4.9 times lower, and the identity jump of the crisis name falls by a factor of 7.4.

These gains have a price. Matching (2) is non-injective in 291 of the 292 refits. The bull template inherits the burn-in's high-volatility, high-return component from early 2021 (Fig. 4). Later fits never produce a component near it, so it receives members in only 35 refits, and under (4) it stays frozen after mid-2022. Calm absorbs the slack and covers 88.5% of bars, and no bar is labelled bull after burn-in. A diagnostic one-to-one Hungarian assignment brings bull back to 5.1% of bars (+11.9 bp), but crisis then grows to 31.3% and its volatility falls to 1.15%. Neither rule alone keeps both coverage and sharp semantics.

B. Trading performance

Table III and Fig. 5 summarise performance out of fold. With W_2 labels the conditioned agent has a Sharpe ratio of −0.16 and loses 29%. Its blind twin does much worse (−1.12, −93%), and buy-and-hold gains 156% with a Sharpe of 0.35. All intervals for the conditioned agent overlap buy-and-hold. The archived moment-sort run reached 0.41, mostly through folds 2 and 3 (Table I), and its interval overlaps ours. By regime (Table IV), the conditioned agent holds a net short position on average in bear (852 bars) and crisis (214 bars) segments (−0.23 and −0.17) while the blind twin stays long (+0.17 and +0.24), and its bear-segment Sharpe ratio of −0.19 is far better than the blind twin's −3.20 and buy-and-hold's −1.94. In calm segments it is long (+0.41) and trails buy-and-hold (Sharpe −0.17 against 0.71).

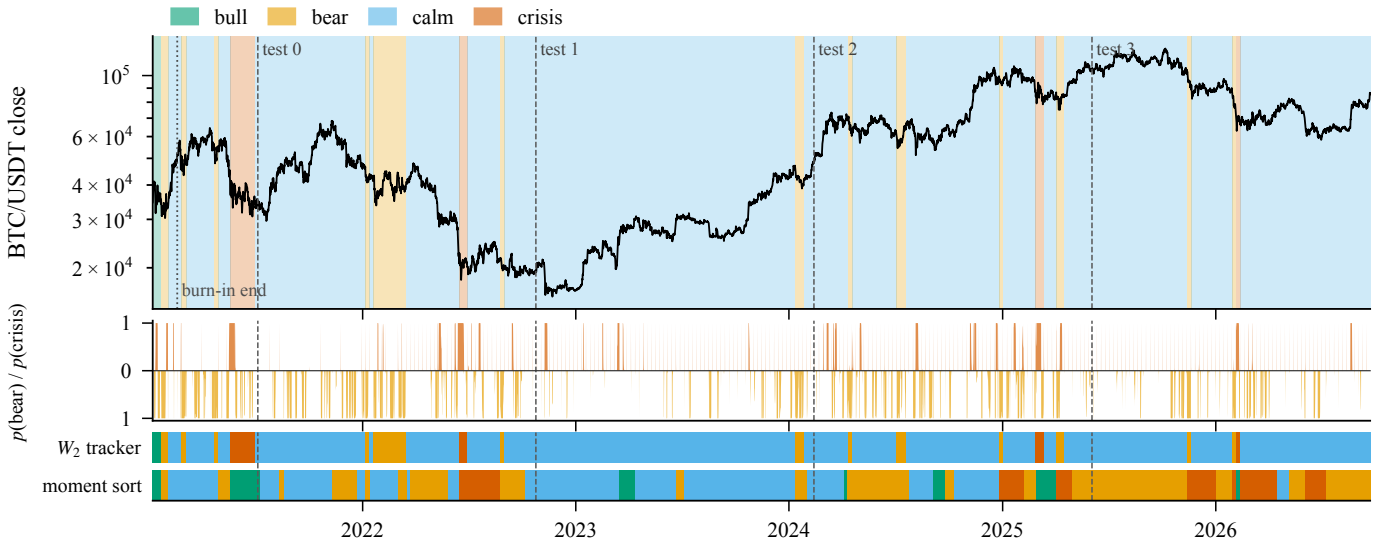


Fig. 3. BTC/USDT four-hour close (log scale) shaded by the dwell-smoothed W_2 tracker label (top). Tracker probabilities of crisis (up) and bear (down) (middle). Labels from the W_2 tracker and from moment sorting on the same 292 refits (bottom). Dashed lines mark the start of each test fold.

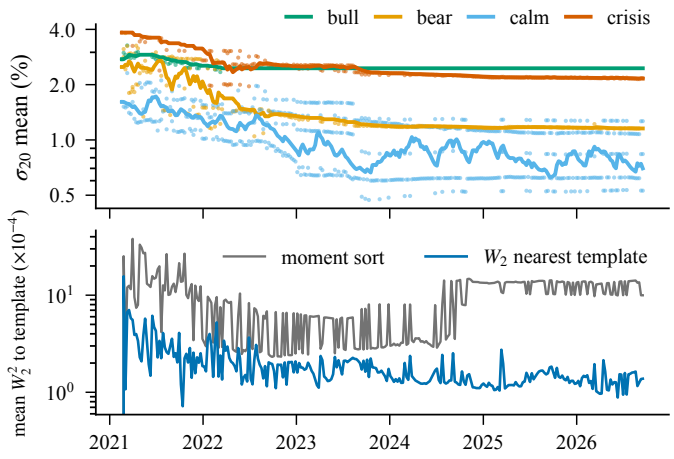


Fig. 4. Top: template volatility (lines) and fitted component volatilities coloured by their W_2 assignment (dots) at each refit. The bull template freezes once it stops receiving members. Bottom: mean squared W_2 distance between components and the template they are named after, for the W_2 rule and for moment sorting.

TABLE III
OUT-OF-FOLD PERFORMANCE OF THE FOUR-HOUR STUDY, 11,430 BARS (95% BOOTSTRAP CI)

System	Sharpe	Sortino	MDD	Return
Conditioned (W_2)	-0.16 [-0.98, 0.63]	-0.22 [-1.34, 0.91]	-77.0%	-29.3%
Blind twin	-1.12 [-1.96, -0.27]	-1.51 [-2.55, -0.37]	-94.4%	-93.0%
Buy-and-hold	0.35 [-0.51, 1.24]	0.50 [-0.71, 1.81]	-77.0%	$+156.2\%$
Cond. (archived)	0.41 [-0.41, 1.20]	0.58 [-0.58, 1.77]	-79.8%	$+139.2\%$

TABLE IV
PER-REGIME SHARPE RATIO AND MEAN POSITION OF THE FOUR-HOUR STUDY (OUT OF FOLD, W_2 LABELS; NO BAR IS LABELLED BULL).

		Sharpe			mean position	
Regime	n	Cond.	Blind	B&H	Cond.	Blind
bear	852	-0.19	-3.20	-1.94	-0.23	$+0.17$
calm	10,364	-0.17	-1.01	0.71	$+0.41$	$+0.25$
crisis	214	0.19	1.22	-2.23	-0.17	$+0.24$

C. Faithfulness

The reference policies behave as designed (Fig. 6): the faithful rule passes on 97.9% of bars, failing ablation only where its output is already within 0.15 of zero, and the ignoring and inverted rules pass on no bar. For the conditioned PPO agent, ablation passes on 60.7% of bars with a median $|\Delta^{\text{ab}}|$ of 0.42, so the regime input is load-bearing. The counterfactual check passes on only 0.9%, and the combined rate is 0.1% (11 of 11,430 bars), against 33.5% for the archived agent.

Fig. 7 explains the failure. Forcing *bear* sends the median position to -1, and the forced-bear position lies below the forced-calm position on 76.5% of bars: the agent has learned a defensive response keyed to bear. Forcing *crisis* gives a median of $+0.64$, defensive relative to calm on only 30.1% of bars, and forcing *bull* gives $+0.53$, below the realised median of $+0.74$. The two directions of the counterfactual check pass on 23.3% (crisis cuts) and 19.9% (bull raises) of eligible bars, and their conjunction almost never holds. The tracker's probabilities are almost one-hot (mean maximum 0.975), crisis occupies 3.8% of bars, and one-hot bull never occurs in training after burn-in, so e_{bull} and e_{crisis} lie far from, or outside, the data the policy was fitted on. The archived labels kept all four regimes populated (bull 6.1%, crisis 12.4% of test bars), and their counterfactuals were in distribution. The counterfactual rank

TABLE V

FIVE-SEED REPETITION OF THE FOUR-HOUR WALK-FORWARD STUDY. ENTRIES ARE THE MEAN ACROSS SEEDS 42–46, WITH THE STANDARD DEVIATION OF THOSE FIVE NUMBERS IN PARENTHESES. RETURN IS THE OUT-OF-FOLD TOTAL RETURN IN PERCENT. ABLATION AND FAITHFULNESS ARE SHARES OF OUT-OF-FOLD BARS, IN PERCENT. FAITHFULNESS IS THE SHARE PASSING ALL THREE CHECKS. THE BLIND TWIN DOES NOT DEPEND ON THE LABELLING. BUY-AND-HOLD IS +156.2% WITH SHARPE 0.35 ON EVERY SEED. THESE NUMBERS DO NOT REPLACE TABLE III.

	conditioned		conditioned			faithfulness	
Labelling	return (%)	Sharpe	pos. s.d.	$\lvert a\rvert > 0.95$	ablation (%)	legacy	v2
Tracker v1	+57.8 (257)	−0.27 (0.78)	0.88 (0.09)	0.79 (0.11)	28.9 (24.8)	0.7 (0.8)	0.6 (0.8)
Tracker v2	−83.6 (17.6)	−0.96 (0.47)	0.89 (0.05)	0.79 (0.09)	27.7 (12.9)	2.4 (2.9)	2.0 (2.6)
Moment-sorted	−44.0 (92.3)	−0.95 (0.96)	0.82 (0.06)	0.71 (0.04)	42.9 (13.8)	12.5 (9.6)	11.5 (9.7)
Blind twin	−87.1 (7.6)	−0.93 (0.36)	0.90 (0.04)	0.82 (0.10)	–	–	–

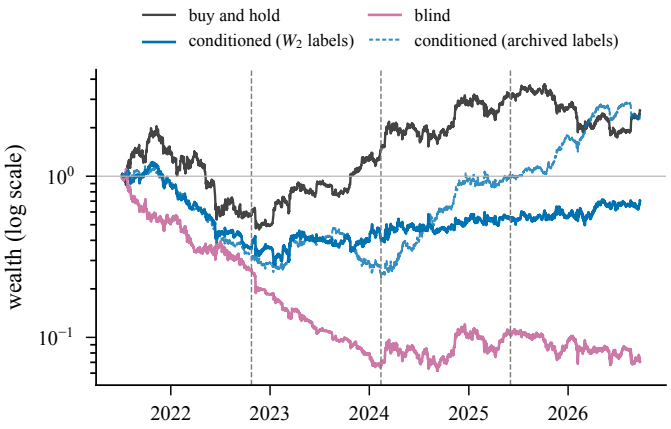


Fig. 5. Out-of-fold wealth (log scale). Dashed vertical lines separate test folds. The dashed blue curve is the archived conditioned agent trained on moment-sorted labels.

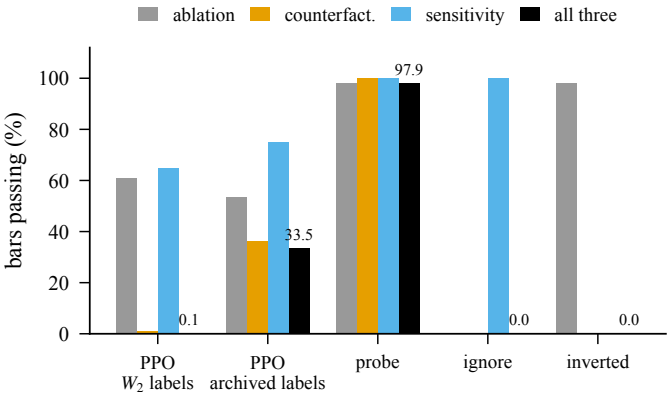


Fig. 6. Share of out-of-fold bars passing each check, for the PPO agent trained on W_2 labels, the archived PPO agent trained on moment-sorted labels, and three reference policies scored on the W_2 probabilities. Numbers give the share passing all three checks.

uses $\arg\max_g p_t(g)$, which agrees with the dwell label on only 78.8% of bars, and a flat sensitivity curve passes the legacy check.

Limit-aware re-score. A counterfactual can also be impossible to pass: an agent within 0.10 of +1 cannot raise its position by 0.10 when bull is forced, and one within 0.10 of −1 cannot cut it when crisis is forced. Re-scoring the published run with

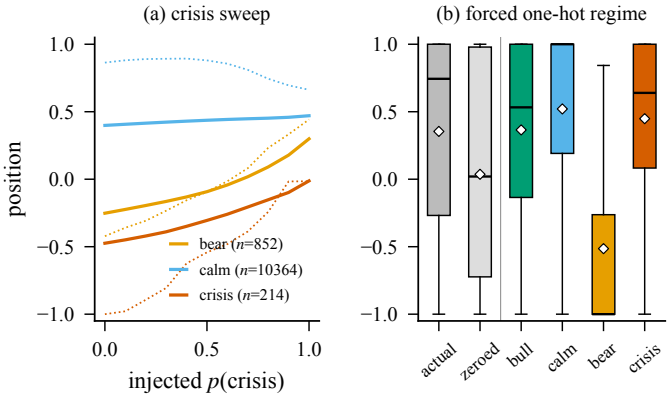


Fig. 7. Interventions on the conditioned agent (W_2 labels). (a) Mean (solid) and median (dotted) position against injected crisis probability, grouped by the bar's label. (b) Distribution of the realised position, the position with the regime input zeroed, and positions under one-hot regime inputs (boxes: interquartile range; diamonds: means).

such bars set aside, 7,040 of the 11,430 bars (61.6%) are untestable. On the 4,390 testable bars the counterfactual check passes on 2.4% and all three checks on 0.25%; the 11 passing bars are the same. This is reported beside the 0.1% headline, not instead of it: removing the position limit as an explanation leaves the faithfulness rate small.

The lesson is methodological. Swapping one well-motivated regime component for another, with architecture, data and training unchanged, improved label semantics and reversed the faithfulness verdict. A verdict cannot be carried over to a modified pipeline.

D. *Five-seed repetition of the four-hour study*

The same walk-forward recipe was repeated for seeds 42–46 on the frozen candles through 22 Sep 2026 12:00 UTC, for tracker v1, tracker v2 and moment-sorted names on the same causal refits, with a circular block bootstrap of block length $\lceil\sqrt{n}\rceil$. Tracker v2 adds template birth and a forward-only posterior; it labels 121 of 12,258 post-burn bars as bull (0.99%, mean +3.3 bp), a real but thin regime.

Table V shows that the single run is not a stable gain. The tracker-v1 conditioned mean return of +57.8% (standard deviation 257) is one seed. Four seeds lose money (−37.5%, −96.3%, −61.6%, −30.8%), and seed 46 returns +515%, of

TABLE VI

FIVE-SEED NATIVE TWINS AFTER THE FROZEN RECIPE. ENTRIES ARE THE MEAN ACROSS SEEDS 42–46, WITH THE STANDARD DEVIATION OF THOSE FIVE NUMBERS IN PARENTHESES. RETURN IS NET OF TURNOVER. "BEATS" IS THE SHARE OF SEEDS WHOSE CONDITIONED LOG-RETURN SUM EXCEEDS THE BLIND TWIN. ABLATION IS THE LEGACY PASS RATE. BUY-AND-HOLD HAS A FIXED POSITION, SO ITS RETURN DOES NOT VARY WITH THE SEED. THE AGENTS ARE NOT REGULARIZED. THIS IS THE MAIN COMPARISON.

		conditioned			blind					
Tape	Window	return (%)	pos. s.d.	Sharpe	return (%)	pos. s.d.	Sharpe	beats	ablation (%)	B&H (%)
BTC 1 h	holdout	2.8 (38.2)	0.55 (0.36)	−0.06 (0.86)	−26.1 (12.8)	0.44 (0.45)	−0.72 (0.44)	0.80	7.3 (13.6)	−23.3
BTC 1 h	live	0.0 (5.6)	0.30 (0.44)	−0.07 (3.12)	1.8 (4.8)	0.44 (0.45)	0.94 (2.66)	0.40	7.5 (14.6)	−3.3
BTC 15 min	holdout	−27.1 (4.9)	0.09 (0.19)	−0.71 (0.15)	−25.7 (2.8)	0.15 (0.33)	−0.66 (0.09)	0.20	0.1 (0.3)	−23.7
BTC 15 min	live	−1.5 (4.8)	0.17 (0.37)	−0.90 (2.68)	−2.1 (3.1)	0.00 (0.00)	−1.22 (1.73)	0.20	0.1 (0.2)	−3.5
ETH 1 h	holdout	−12.8 (56.4)	0.74 (0.22)	−0.52 (1.11)	−19.0 (39.6)	0.02 (0.04)	−0.47 (0.65)	0.20	25.6 (20.0)	−34.1
ETH 1 h	live	0.4 (9.3)	0.53 (0.14)	−0.03 (4.14)	−5.5 (8.3)	0.02 (0.05)	−2.68 (3.71)	0.80	29.5 (30.2)	−8.5
ETH 15 min	holdout	−38.4 (8.8)	0.36 (0.29)	−0.79 (0.23)	−7.7 (44.5)	0.37 (0.37)	−0.30 (0.84)	0.40	0.5 (0.5)	−34.7
ETH 15 min	live	−5.4 (3.2)	0.33 (0.43)	−2.55 (1.54)	0.6 (8.9)	0.32 (0.44)	0.11 (4.08)	0.40	0.2 (0.2)	−8.8

which fold 2 alone returns $+870\%$ against $+111\%$ for buy-and-hold on that fold. The mean Sharpe ratio stays negative at -0.27. Seed 42 returns -37.5% with Sharpe -0.22, next to the published -29.3% and -0.16; the frozen file differs from the published run by one restored close. Tracker v2 loses on every seed, and moment-sorted names lose on four seeds. The legacy all-check rate is 0.7% for tracker v1, 2.4% for tracker v2 and 12.5% for moment-sorted names, and verifier v2 sits at or below it in each case. The rail share of 0.71–0.82 is lower than for the native twins but still a policy that lives on the rails.

E. Five-seed native twins

The native recipe was chosen by a one-factor grid on the validation split only (seed 42, both assets and timeframes). The factors were the reward (raw, excess over buy-and-hold, differential Sharpe), the window sampler (uniform, regime-balanced, return-balanced), the entropy coefficient (0.01, 0.02), a per-episode perturbation of the regime probabilities (0, 0.05, 0.1) and the tracker (v1, v2). The pass bar was fixed in advance and did not use returns: on all four tapes, both twins with position standard deviation above 0.05, both with a share of bars at $|a| > 0.95$ below 0.80, and ablation on at least 10% of bars. A cell's score is the mean over tapes of the summed relative shortfalls from those bars. No cell passed; every validation twin spent more than 90% of bars at $|a| > 0.95$. The baseline had the lowest mean deficit (about 0.985, against 1.33 for the next cell and 2.65 for excess-over-buy-and-hold), so `configs/final.yaml` stayed the recipe: raw reward, scale 100, entropy coefficient 0.01, uniform 2,000-bar windows, tracker v1, no perturbation and no faithfulness regularizer. Holdout and live were then scored once on seeds 42–46 and were not used to change it.

Table VI is the comparison, and Figs. 8 and 9 draw each holdout and live path. Buy-and-hold lost money on every holdout and live tape. Every mean holdout Sharpe ratio is negative for both twins. The conditioned twin beats the blind twin on a majority of seeds in two of eight windows only: BTC 1 h holdout ($+2.8\%$ against -26.1%, beats share 0.80, ablation 7.3%) and ETH 1 h live ($+0.4\%$ against -5.5%, beats share 0.80, ablation 29.5%). On ETH 1 h holdout the conditioned mean is one large path (beats share 0.20). It does not win overall. The mean share of bars at $|a| > 0.95$ runs from 0.971 to 1.000 across the sixteen twin–window means, so every twin spends 97–100% of bars on the rails, and a large position standard deviation in Table VI reflects seeds that disagree about which rail to hold. The ETH 1 h blind twin is flat on both windows (position standard deviation 0.02), and the BTC 15-min live blind twin has standard deviation 0 on all five seeds. The median absolute ablation effect is 0 on every tape except ETH 1 h live. Both verifiers give small rates (Table VII): the largest legacy mean is 0.046, on ETH 1 h holdout, where verifier v2 gives 0.036, v2 is at or below legacy on every row, and no bar was untestable.

TABLE VII

MEAN SHARE OF BARS PASSING ALL THREE FAITHFULNESS CHECKS, ACROSS SEEDS 42–46, WITH THE STANDARD DEVIATION OF THE FIVE RATES IN PARENTHESES. LEGACY IS THE ONE-HOT VERIFIER. VERIFIER V2 USES TRAINING-SET PROBABILITY VECTORS AND REJECTS A FLAT SENSITIVITY CURVE. THE AGENTS ARE NOT REGULARIZED.

Tape	Window	Legacy	Verifier v2
BTC 1 h	holdout	0.006 (0.008)	0.005 (0.007)
BTC 1 h	live	0.001 (0.002)	0.001 (0.002)
BTC 15 min	holdout	0.001 (0.003)	0.001 (0.003)
BTC 15 min	live	0.001 (0.002)	0.001 (0.002)
ETH 1 h	holdout	0.046 (0.070)	0.036 (0.067)
ETH 1 h	live	0.035 (0.041)	0.021 (0.041)
ETH 15 min	holdout	0.003 (0.006)	0.003 (0.006)
ETH 15 min	live	0.001 (0.002)	0.001 (0.002)

F. Clipping, action heads and a risk-penalised reward

The validation weights for the native recipe were not kept, so seed 42 was retrained on the validation split with that recipe, and the raw Gaussian mean u was recorded on every validation bar before the clip to $[-1, 1]$. On every tape and both twins u lies outside $[-1, 1]$ on 95.6–99.8% of bars, and the executed position lies beyond 0.95 on 95.8–99.8% (Table VIII). The rule fixed before the table was read treats clipping as the cause if at least six of the eight rows have both shares at or above 0.50 and 0.80; all eight rows clear both bars. The policy asks for a position outside the legal interval, and the clip leaves it on the rail.

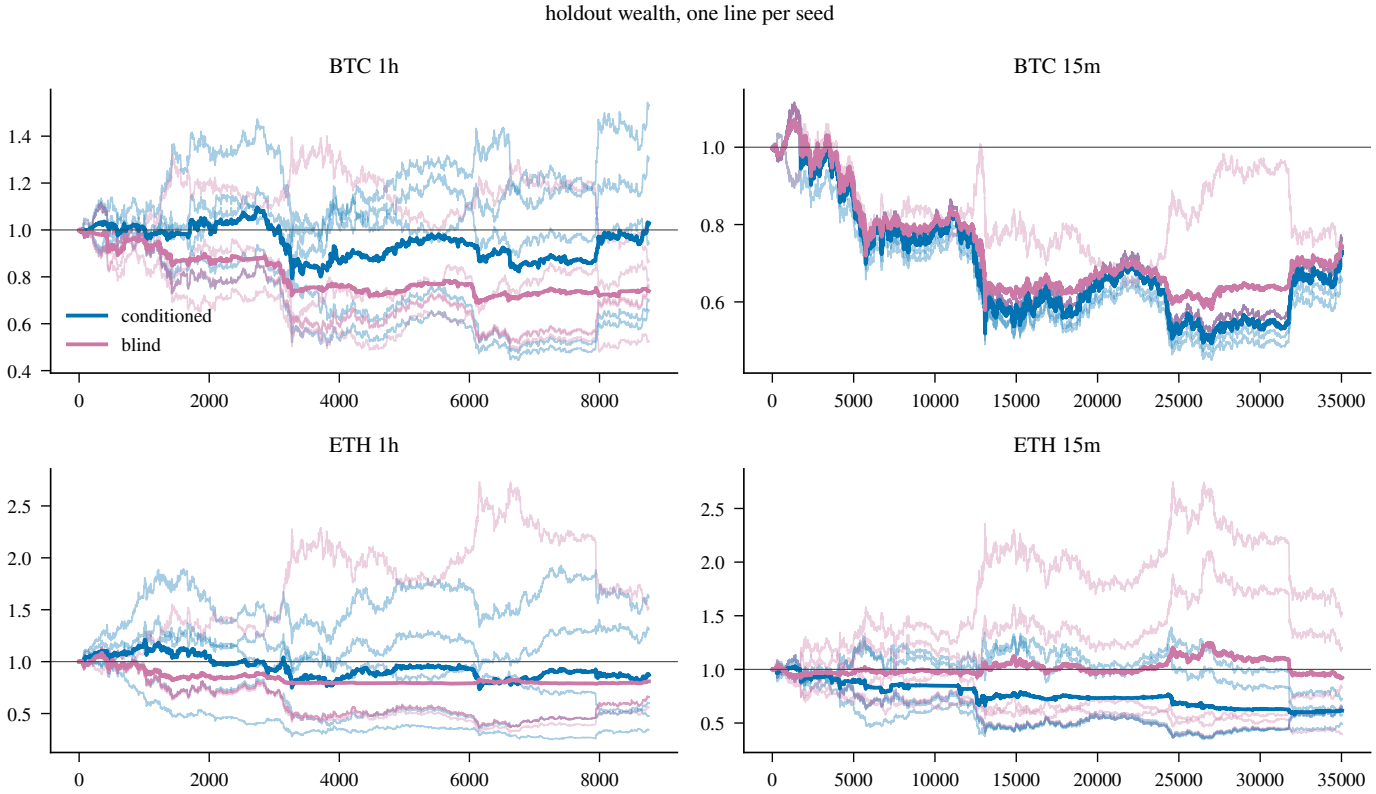


Fig. 8. Holdout wealth for the five frozen seeds. Faint lines are seeds. The heavy line is the mean. Blue is the conditioned twin and pink is the blind twin. The horizontal line is a wealth of 1.

TABLE VIII
VALIDATION-SPLIT SATURATION OF THE NATIVE TWINS, SEED 42, RETRAINED BECAUSE THE VALIDATION WEIGHTS HAD NOT BEEN KEPT. SHARE OF BARS WHOSE RAW GAUSSIAN MEAN HAS ABSOLUTE VALUE ABOVE 1, SHARE WHOSE EXECUTED POSITION HAS ABSOLUTE VALUE ABOVE 0.95, AND THE MEAN ABSOLUTE CHANGE IN THE EXECUTED POSITION.

Asset	Tape	Twin	$\lvert u\rvert > 1$	$\lvert a\rvert > 0.95$	mean $\lvert\Delta a\rvert$
BTC	1 h	cond.	0.956	0.958	0.062
BTC	1 h	blind	0.989	0.990	0.006
BTC	15 min	cond.	0.998	0.998	0.013
BTC	15 min	blind	0.998	0.998	0.003
ETH	1 h	cond.	0.980	0.982	0.030
ETH	1 h	blind	0.990	0.990	0.020
ETH	15 min	cond.	0.995	0.995	0.004
ETH	15 min	blind	0.995	0.996	0.005

Screens. Three validation screens followed, each at seed 42 with the clipped Gaussian under the raw reward as the control and the reward, budget and both twins' head otherwise unchanged. The first replaced the head: `tanh_env` executes $a = \tanh(u)$, and `discrete5` chooses among $\{-1, -0.5, 0, 0.5, 1\}$ with the verifier scoring the expected position. The second and third kept the clipped Gaussian or the five-level head and replaced (5) by a mean-variance reward,

$$R_t^{\text{mv}} = a_t\, r_{t+1} - c\,|a_t - a_{t-1}| - \lambda\, a_t^2 \sigma_t^2, \qquad \lambda \in \{2, 5, 10\}, \tag{10}$$

with σ_t the 20-bar volatility known at t. A reward linear in the position is maximised at a rail whenever the expected return is non-zero. Under (10) the expected reward, ignoring turnover, is maximised at $a^* = \mathrm{E}[r_{t+1}]/(2\lambda\sigma_t^2)$, an interior position unless the signal is large relative to risk, so this was the principled test of the rail mechanism. The pass bar was fixed before each screen and is the grid bar of Section VI-E: on all four tapes, both twins with position standard deviation above 0.05, both below 0.80 at $|a| > 0.95$, and conditioned ablation on at least 10% of bars. Returns were not an input. Each screen was validation-only, and holdout and live were not scored for any of these cells.

No cell passed (Table IX). Under the clipped Gaussian the penalty did not move the policies off the rails: at every λ, every twin on every tape is at $|a| > 0.95$ on at least 88.9% of bars. Under the five-level head the penalty did remove the rails. Every twin is at ± 1 on at most 5.9% of bars, except the ETH 15-min blind twin at $\lambda = 2$ (95.9%). The policies instead collapsed to a single level, usually 0 or $+0.5$. The number of twins holding one level on more than 90% of bars is 4, 3 and 6 of 8 at $\lambda = 2$, 5 and 10, and at $\lambda = 10$ three twins hold position 0 on every bar. The number of twins at or below position standard deviation 0.05 rises from 1 of 8 at $\lambda = 2$ to 3 of 8 at $\lambda = 5$ and $\lambda = 10$. A larger λ made the collapse worse, not better.

Interpretation. A hypothesis consistent with all three screens is that the policies behave as if the features carry little

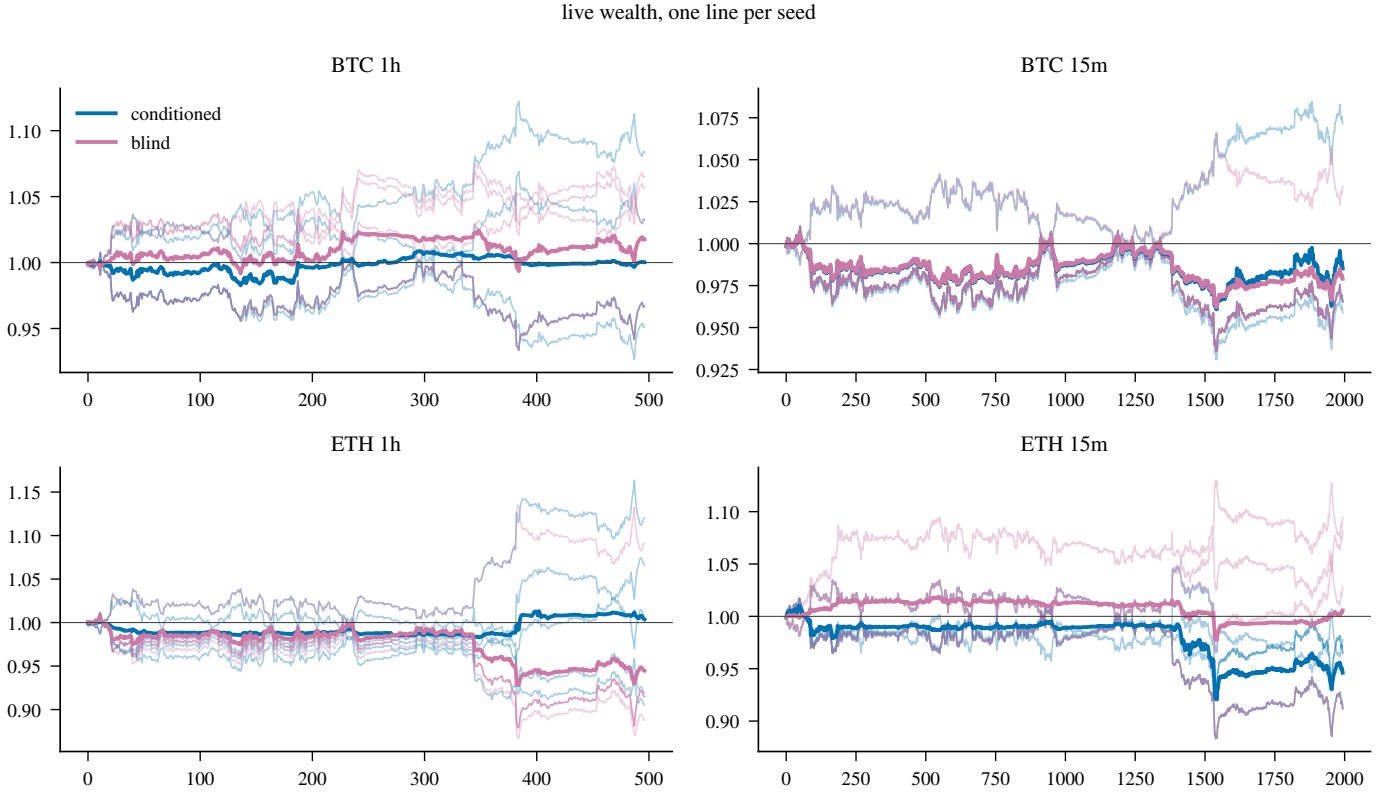


Fig. 9. Live-window wealth for the same five seeds. The scale is the three weeks after 22 Sep 2026, not the holdout year. Faint lines are seeds and the heavy line is the mean.

TABLE IX
VALIDATION SCREENS OF ACTION HEADS AND REWARDS, SEED 42. PASS: TAPES ON WHICH BOTH TWINS MEET THE BAR, OUT OF 4. $|a|>0.95$: LARGEST SHARE OVER THE EIGHT TWINS (FOR `discrete5`, THE EXECUTED LEVEL IS ± 1). ABL.: SMALLEST CONDITIONED ABLATION RATE OVER THE FOUR TAPES. RAILS: SHARE AT $|a| > 0.95$ OF AT LEAST 0.80. MV: MEAN-VARIANCE REWARD (10). NO CELL WAS SELECTED.

Head, reward	Pass	$\lvert a\rvert>0.95$	Abl.	Binding failure
Gaussian clip, raw	0/4	0.998	0.008	rails on all tapes
`tanh_env`, raw	0/4	1.000	0.000	rails on all tapes; BTC cond. near-constant
`discrete5`, raw	3/4	0.712	0.000	BTC 1 h cond. constant 0
Gauss. clip, MV λ=2	0/4	1.000	0.006	rails on all tapes
Gauss. clip, MV λ=5	0/4	1.000	0.000	rails; BTC 1 h cond. constant $+1$
Gauss. clip, MV λ=10	0/4	1.000	0.000	rails; BTC 15 min twins constant $+1$
`discrete5`, MV λ=2	1/4	0.959	0.025	ablation $<$ 0.10 on 3 tapes
`discrete5`, MV λ=5	0/4	0.001	0.050	blind s.d. $\leq$ 0.05 on 3 tapes
`discrete5`, MV λ=10	1/4	0.000	0.000	constant 0: BTC 1 h cond., ETH 1 h both

predictable next-bar signal. The best they then find is a fixed position: at a rail under the linear reward, where any small mean return favours a corner, and at an interior level once risk is penalised, where a^* shrinks towards 0 as λ grows. On this reading the rails are a symptom of a weak signal, not its cause, which would explain why changing the head or the reward moved the policy's resting point without making it responsive. A direct test of feature predictability was not run, so this remains a hypothesis.

VII. LIMITATIONS AND FUTURE WORK

Within-window smoothing. Each HMM is fitted only on earlier rows, but Viterbi and forward–backward decoding over the 42-bar window let a bar's probability condition on up to 41 later bars of that window. Labels are never revised after their window closes, but they are not strictly filtered; per-regime results carry this caveat, and the native tapes use the same decoder.

Template starvation and counterfactuals. Non-injective matching with a frozen unmatched template removes the bull regime, and one-to-one matching is not a complete fix (Sec. VI-A). Tracker v2 restores a thin bull regime but loses on every seed (Sec. VI-D). One-hot interventions test the four-hour policy off its training distribution; verifier v2 draws in-distribution vectors and still reports small rates. No agent was trained with a faithfulness penalty.

Statistics and live evidence. The i.i.d. bootstrap of Tables I and III ignores serial dependence, so a block bootstrap would widen those intervals. The live window is 21 days, in which buy-and-hold lost money on every tape; it can show the direction of a gap but cannot establish one. Paper fills at the

candle close with a flat 10-bp cost ignore slippage, latency and funding. The thresholds δ_{ab} and δ_{cf} and the ordering ρ are design choices.

Position rails. No action head or reward tested here produced a policy that uses the interior of $[-1, 1]$ while staying responsive: the clipped Gaussian and the tanh head stay on the rails, and the five-level head with or without a mean-variance penalty settles on a fixed level on some tapes. The untested next step is a direct predictability test of the features for the next-bar return, and richer inputs if that test finds little signal.

VIII. Conclusion

We built a regime-conditioned RL trading agent with a Wasserstein identity tracker and a suite that intervenes on the regime input of every decision. On BTC/USDT four-hour data the tracker kept a stable volatility ordering across 292 causal refits and starved the bull template. The conditioned agent used those labels, since removing them changed its position on 60.7% of the 11,430 out-of-fold bars and it reduced exposure in bear and crisis segments, yet only 0.1% of bars passed all three checks (0.25% of the bars the position limit leaves testable), against 33.5% for moment-sorted labels. Its out-of-fold Sharpe ratio was -0.16, against -1.12 for the blind twin and 0.35 for buy-and-hold, and a five-seed repetition shows that the conditioned mean return of $+57.8\%$ is one seed at $+515\%$, with a mean Sharpe ratio of -0.27.

Five seeds of timeframe-native twins with a recipe frozen on validation give a negative mean holdout Sharpe ratio for both twins, and the conditioned twin beats the blind twin on a majority of seeds in two of eight windows only. Those twins hold $|a| > 0.95$ on 97–100% of bars. Neither a tanh or five-level action head nor a mean-variance reward produced a policy that uses the interior of $[-1, 1]$ while staying responsive: the penalty either left the rails in place or replaced them with a fixed interior level. We read this, as a hypothesis not yet tested, as a sign that the features carry little predictable next-bar signal. A faithfulness verdict has to be recomputed whenever the regime layer or the training recipe changes.

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